🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 6 · STANDARD 6.AT.7

Generate Equivalent Expressions

Lesson 6-8 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes

Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can use the commutative, associative, and identity properties to rewrite expressions.
Language Objective: I can explain my reasoning using the words commutative property, associative property, and identity property.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

- Properties of Operations
- Commutative Property
- Associative Property
- Identity Property
- Property
- Distributive Property

👆 *Tap any word to see what it means and a picture.*

1. Rules that let you reorder or regroup without changing the ______________.

2. The rule that order does not change a sum or product, like a + b = b + a, is the ______________.

3. The rule that grouping does not change a sum or product, like (a + b) + c = a + (b + c), is the ______________.

4. The rule that adding 0 or multiplying by 1 keeps a number the same is the ______________.

5. A rule that is always true in math is a ______________.

6 The rule that lets you multiply a number across a sum, like $a(b + c) = a \times b + a \times c$, is the ______.

Write About the Math

What properties did the engineer use, and why does rearranging make the calculation easier?

¿Qué propiedades usó la ingeniera, y por qué reacomodar hace más fácil el cálculo?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Properties of Operations** *Propiedades de las operaciones*
- ☐ **Commutative Property** *Propiedad conmutativa*
- ☐ **Associative Property** *Propiedad asociativa*
- ☐ **Identity Property** *Propiedad de identidad*
- ☐ **Property** *Propiedad*

Model: *The commutative property lets you change the order of a sum without changing the total.* — Students name the commutative property and explain that 3 + 5 + 2 gives 10 in any order because reordering addends does not change the sum.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The engineer used the ____ property to change the ____ and the ____ property to change the ____. This makes it easier because 4 × 25 = ____, which is a friendly number to multiply by.**
- **My answer is ____ because ____.**

  Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**

  Mi afirmación es ____.
- **My evidence is ____, so ____.**

Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Properties of Operations
2. **Notes 2:** Commutative Property
3. **Notes 3:** Associative Property
4. **Notes 4:** Identity Property
5. **Notes 5:** Property
6. **Notes 6:** Distributive Property
7. **Watch for this mistake:** *A common mistake in Properties of Operations is applying the commutative property where it does not belong, or confusing it with the associative property. Students often assume 15 − 8 = 8 − 15 (it does not: 7 ≠ −7 — commutative only works for addition and multiplication), or they see parentheses move and call it "associative" even when the numbers inside just swapped places, like (2 + 3) + 4 = (3 + 2) + 4 (that is commutative — the order inside the parentheses changed, not the grouping). Before you label a property, ask: did only the order change (commutative), did only the grouping change (associative), or was a 0 or 1 involved (identity)?*
8. **Try It 1:** A. Associative Property — *The grouping (parentheses) changed but the order of the numbers stayed the same, so this is the Associative Property.*
9. **Try It 2:** A. Identity Property — *Multiplying by 1 keeps the same value, so this is the Identity Property of Multiplication.*